🔥 Hearth

Dignified, confidential cash aid on a public ledger.

Donors see a provable total. An auditor reconciles every unit.
Nobody can read what any family received.

- Live on the Tari Ootle Esmeralda testnet · 20 tests · MIT

The problem: transparency can hurt the people aid is for

- Cash grants are sized by household size, disability, illness, protection status.
- On a transparent chain, a grant amount is sensitive personal data, published forever, readable by neighbours, landlords, armed groups and data brokers.
- The ICRC data-protection handbook and GDPR Art. 9 say: don't. "Do No Harm."
- Yet donors need proof: no double-dipping, no ghost households, books that reconcile.

Today agencies must choose: **transparency *or* safety.**

Hearth: each party sees exactly what it needs

Who	Sees
👁️ Public / armed group	Funds raised, household count, grants per round. No amounts.
🔑 Each household	Its own grant plus a private note, opened with its own key
📜 Donor's auditor	Every amount + treasury, reconciled to the unit, with no spending power
🏘️ Another household	Only its own grant
🦹 Double-dipper	Rejected on-chain via salted-hash enrolment, with no PII stored

Built for the field, not the demo

- **Salted-hash enrolment:** raw IDs are refused; each tag is unique forever, even after offboarding.
- **Lost phone:** `replace_wallet` keeps the household's place and paid status, so a new wallet never creates a second entitlement.
- **Atomic rounds:** up to 64 grants in one tx. One bad grant rejects all of them. The only event is a *count*.
- **Can't leak by mistake:** revealed grants are rejected. The auditor key is locked on the token for its lifetime.
- **Proof to a landlord:** a family can prove one grant and reveal nothing else.

Live on the public testnet, verified from chain

One transaction: **enrol ×3** → **open_round** → **disburse_round**

`fe39987741d91b97046e516e47806fd89ac8e644fd12525e1a664423fbe94d2b`

- Household keys open 3000 / 1500 / 2000, each only its own.
- Auditor: 3000 + 1500 + 2000 + treasury 3500 = 10,000 = funds raised ✔
- Re-enrolling Household A under a 2nd wallet: "Household is already enrolled (duplicate enrolment tag)"

Template `28d53582…4d89e05` · Explorer: explorer.tari.mw

Path to a real pilot

- **Voucher mode:** spend at vetted merchants; the auditor reconciles redemptions.
- **Governance:** multi-sig operator; per-donor auditor keys with rotation.
- **Reach:** a USSD/SMS front end for feature phones.
- **Pilot:** one local NGO, one district, transfer values sized by the Sphere minimum standards.

Honest limits are in THREAT_MODEL.md: the chain proves *uniqueness*, not that a person is real, so field verification stays essential.